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Data 612 Recommender System – Summer 2025 Project 3

Understanding Latent Space Dimensions

Thank you for your feedback. I wanted to offer more insight into my decision to visualize the latent space and how I tackled the dimensional aspect. Early in the course, I was honestly worried about my recommender's performance, especially since skincare recommendations can be so nuanced for melanin-rich users. I needed a way to check that the system was truly learning something meaningful, not just churning out numbers.

I was curious about how others approached the hidden relationships between products. I found a Kaggle notebook where SVD was used to build embeddings that revealed product similarities. This discovery motivated me to delve into latent factors and their connection to user-product interactions.

For my project, I implemented singular value decomposition (SVD) to reduce the dimensions of the user-item matrix. I experimented with several values of k , starting from 10, closely monitoring both reconstruction error and the interpretability of resulting plots. Ultimately, I chose some dimensions that struck a good balance between retaining significant variance and keeping the visuals clear. I then took those embeddings and ran PCA to project the latent space onto two dimensions for visualization.

The resulting plots let me see which products clustered together, highlighting whether my system was capturing fundamental, meaningful similarities, not just based on shared ratings, but also on underlying product attributes relevant to my audience, like being safe for sensitive skin or popular among users with deeper skin tones.

Sharing this work in class, even without a fully updated RMarkdown report, gave me a chance to reflect on the value of embedding models and dimensionality reduction. In the future, I intend to use these techniques in my final project to further evaluate and refine the fairness and quality of recommendations across several types of recommender systems.